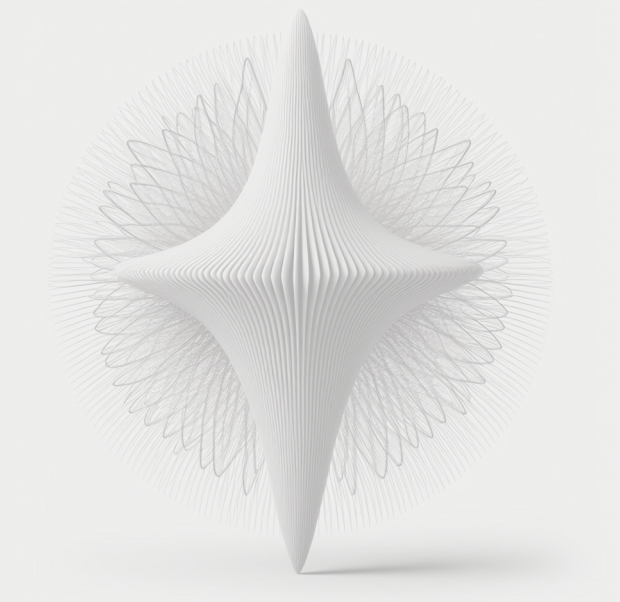


Signals and Systems

Lakshmi Narasimhan
IIT Madras

Disclaimer: There could be errors and typos in these slides. Use with caution.

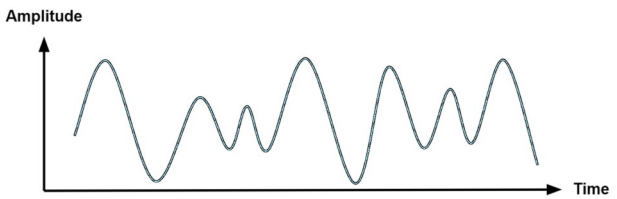
Outline of the Course

- Introduction
- One dimensional signals
- SISO systems
  - Linear and time invariant (LTI) systems
- Fourier analysis
- Laplace analysis

Introduction

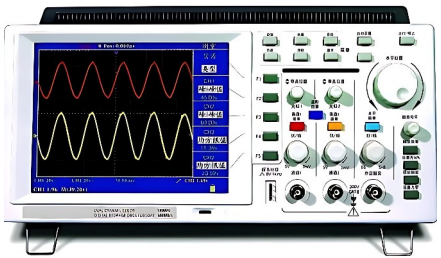
Signals

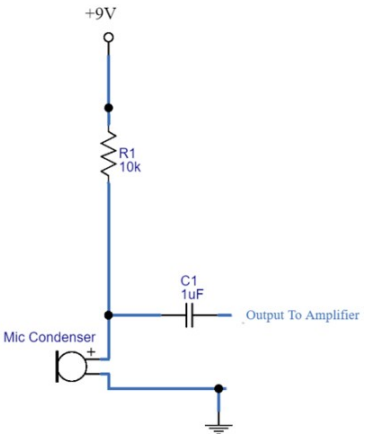
- A signal is a function *f(x)*
  - *x* is the independent variable
  - *f()* is the dependent value
  - Can visualize the signal through a plot
- Examples of independent variables
  - Time, frequency, position, etc.
- Examples of dependent value
  - Amplitude, phase, etc.



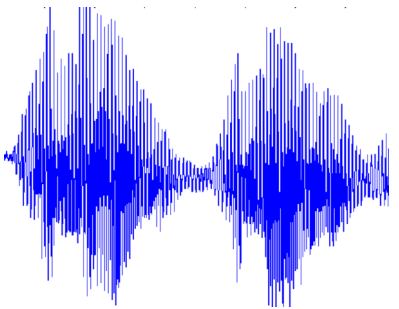
$$f : \mathbb{R} \rightarrow \mathbb{C}$$

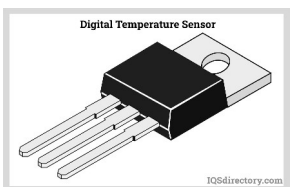
Examples of Signals



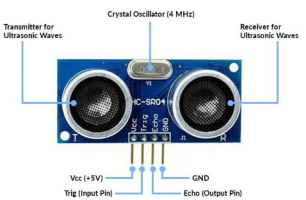


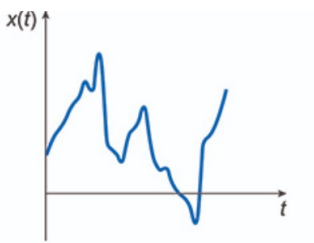
Source: www.circuits-diy.com



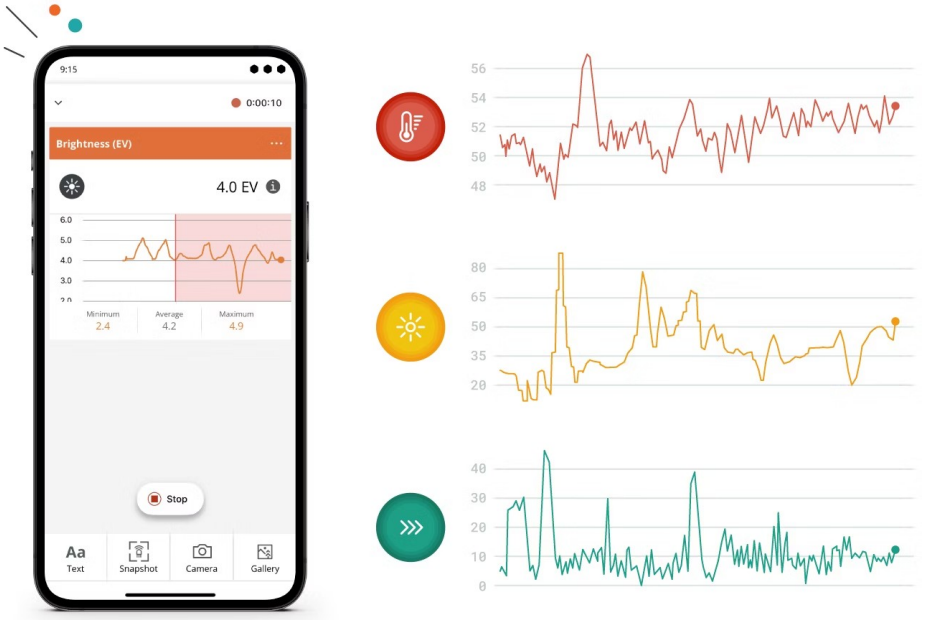






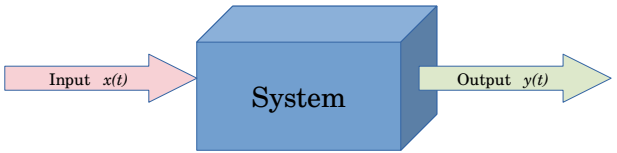


Arduino Science Journal App

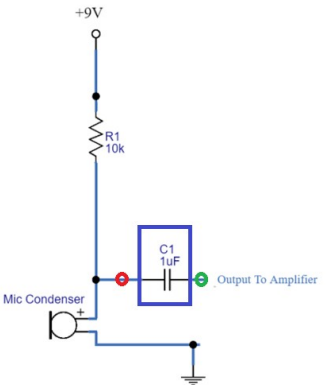


Systems

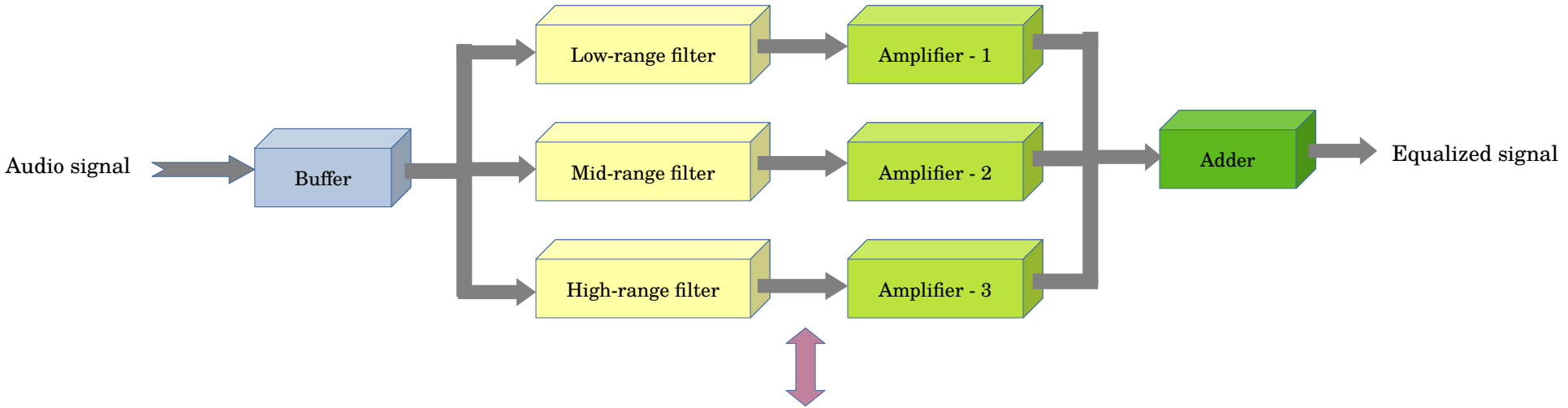
- A *box* that processes an input signal and produces an output signal



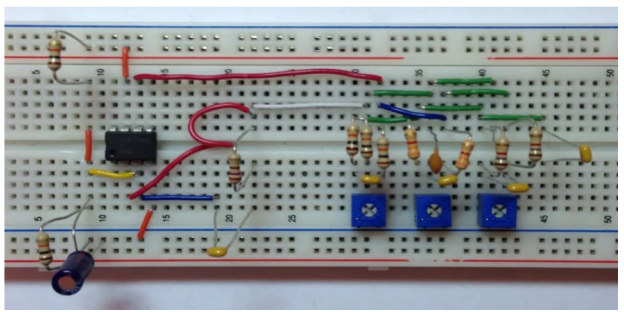
- Example
  - Input: a voltage signal
  - Output: signal without DC component



Example: Building an Audio Equalizer



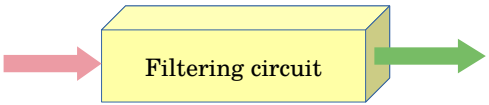
- Each building block is a system with input & output signals
- Parallel and serial connections

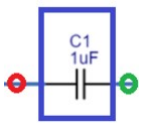


- Each system can be designed, analyzed and optimized separately

Example: Building a Filtering Circuit

- Signal analysis
  - How to model a signal and its frequency components?
  - How to model the signals that are transformed by a system?





- System analysis and design
  - How to model the transformation caused by the system to any signal?
  - How to design the system to provide the desired output?
  - What are the properties/characteristics of this system?

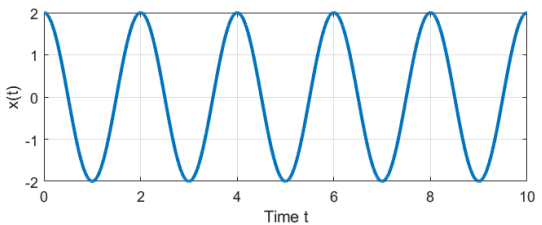
Signal Analysis

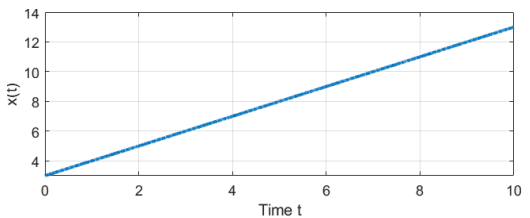
Analytical Representation

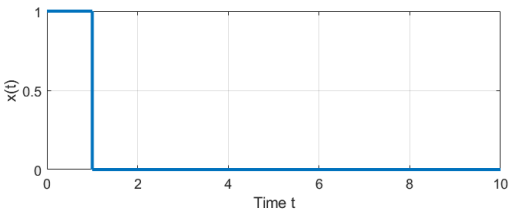
- Mathematical representation – $f : \mathbb{R} \rightarrow \mathbb{C}$
  - Mapping of an independent variable (referred to as time) to a dependent variable (referred to as amplitude)

$$x(t) = 2\cos(\pi t) \qquad x(t) = t + 3 \qquad x(t) = \begin{cases} 1 & \text{if } \ 0 < t < 1 \\ 0 & \text{otherwise} \end{cases}$$

- Graphical representation

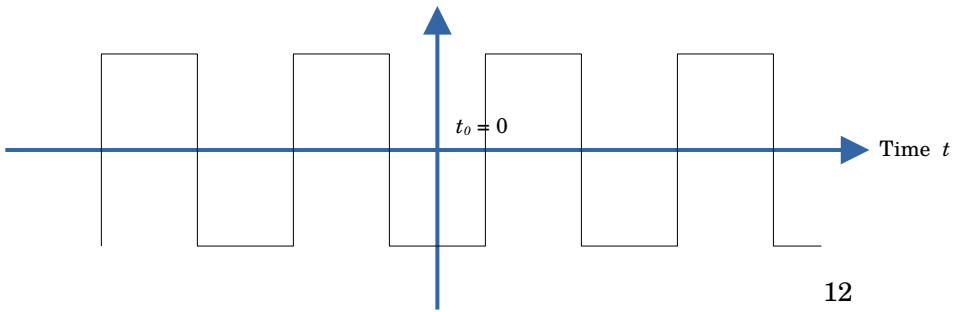






Negative Time

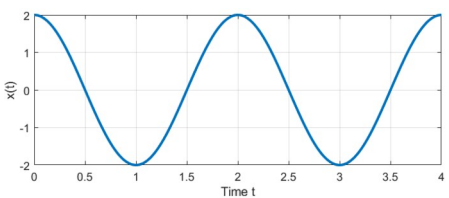
- A mathematical representation of time before the present time
- If a particular system's input or output is observed starting from time t_0, then this starting time (origin) is set as $t = 0$
  - The values of the signal before the start of observation are indicated with negative values of time
  - Examples: $x(t) = 2\cos(\pi t)$ $\quad x(t) = t + 3$ $\quad x(t) = \begin{cases} 1 & \text{if } 0 < t < 1 \\ 0 & \text{otherwise} \end{cases}$



Classification of Signals

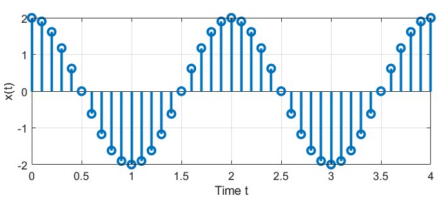
Continuous-time

- The values of the independent variable (t) is from a continuum
- Examples: audio signals, voltage/current signals from a circuit



Discrete-time

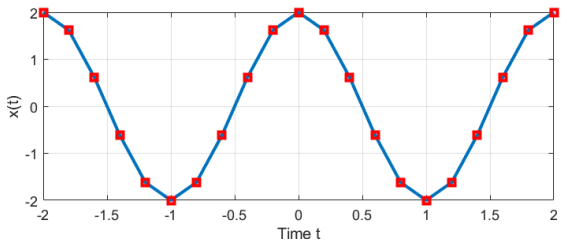
- The independent variable takes only discrete values
- Examples: A sampled signal, everyday's temperature



Classification of Signals

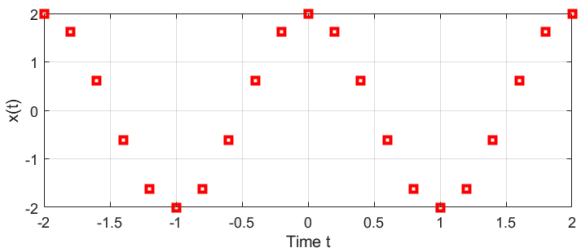
Continuous-time

- Infinitely many time instances between any time interval (t_0, t_1)
- A well-defined function, e.g., $x(t)$
- Amplitude can be from a continuum
- Can be obtained by interpolating a discrete-time signal



Discrete-time

- Finite number of time samples between two time instances
- A sequence, e.g., $x[n]$
- Amplitude can be from a continuum
- Can be obtained by sampling a continuous-time signal $x[n] = x(nT)$



Analog vs Digital

Analog

- Continuous-valued continuous-time
- The amplitude can take values from a continuum
- Analog signals cannot be stored*, but generated and processed by analog circuits

* Inefficient medium such as magnetic tapes, etc., can be used for storage

Digital

- Discrete-valued discrete-time
- The discrete sequence $x[n]$ is further discretized (quantized) in amplitude
- Only digital signals can be stored and processed by a computer (i.e., digital circuits)

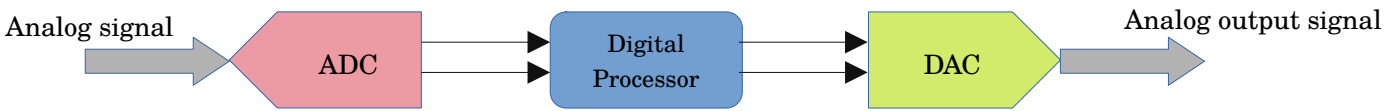
Analog vs Digital

Analog

- All real-life signals (carried in wired or wireless media) are analog

Digital

- All signals processed in a computer or micro-controller are digital



Examples of Signals

Continuous-Time Sinusoid

- $x(t) = A\cos(\omega_0 t + \phi) = A\cos(2\pi f_0 t + \phi)$
  - Amplitude: -A to + A
  - Angular frequency: w_0 radian/second
  - Rotational/linear frequency: $2\pi f_0 = w_0$
  - Period: $T_0 = 1/f_0 = 2\pi/w_0$
  - Starting phase: ϕ
  - Sound from a buzzer

Step and Ramp

- $\mathrm{u(t)} = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{otherwise} \end{cases}$
  - Constant value from the time of observation
  - For CT, Mathematically, discontinuous at t = 0; (converges to ½ at t = 0)
  - Voltage across a battery
- $\mathrm{r(t)} = \begin{cases} t & \text{if } t \geq 0 \\ 0 & \text{otherwise} \end{cases}$
  - Increasing with a fixed slope
  - Relation: $\mathrm{r(t)} = \int_0^t u(\tau) d\tau$
  - Current in an inductor

Rectangular and Triangular Pulse

- $\text{rect}(t) = \begin{cases} 1 & \text{if } -\frac{1}{2} \le t \le \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$ $\quad rect(ta) = \begin{cases} 0 & |t| > 1/2a \\ 1/2 & |t| = 1/2a \\ 1 & |t| < 1/2a \end{cases}$
  - Constant value in an interval
  - Can be obtained as a combination of step signals
  - Voltage signals from a serial/USB port

- $\text{tri}(t) = \begin{cases} 1 - |t| & \text{if } -1 \le t \le 1 \\ 0 & \text{otherwise} \end{cases}$
  - Can be obtained by integrating some combination of rectangular pulses

Examples of Discrete-Time Signals

Discrete-Time Sinusoid

- $x[n] = A\cos(\omega_0 n + \phi) = A\cos(2\pi f_0 n + \phi)$
  - Amplitude: -A to + A
  - Is it always periodic?
    - Only if w_0 is a rational multiple of 2π
    - Linear frequency: f_0 - only if it is rational
  - Period: N, where $f_0 = m/N$ is an irreducible rational number
  - Starting phase: ϕ

Step and Ramp

- $\mathrm{u[n]} = \begin{cases} 1 & \text{if } n \geq 0 \\ 0 & \text{otherwise} \end{cases}$
  - Constant value from the time of observation
  - Scaled variations: $a\mathrm{u[n]}$

- $\mathrm{r[n]} = \begin{cases} n & \text{if } n \geq 0 \\ 0 & \text{otherwise} \end{cases}$
  - Increasing with a fixed slope
  - Relation: $\mathrm{r[n]} = \sum_{k=1}^{n} u[k]$

Rectangular and Triangular Pulse

- $\text{rect[n]} = \begin{cases} 1 & \text{if } -\frac{N}{2} \leq n \leq \frac{N}{2} \\ 0 & \text{otherwise} \end{cases}$
  - Constant value in an interval
  - Can be obtained as a combination of step signals
  - Non-zero duration of N samples

- $\text{tri[n]} = \begin{cases} N - |n| & \text{if } -N \leq n \leq N \\ 0 & \text{otherwise} \end{cases}$
  - Can be obtained by summing some combination of rectangular pulses
  - Non-zero duration of $2N$ samples

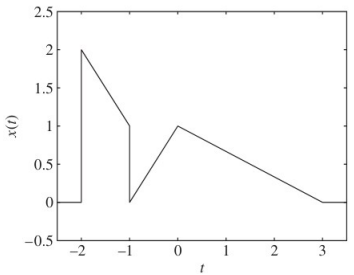
Manipulating Signals

Time Shifting

- Delay or advance a signal in time: $x(t - T)$ or $x(t + T)$
  - $x[n - N]$ or $x[n + N]$
  - Moves the signal to the right or left in time
- Examples
  - TV transmission delays
  - Audio timing adjustment

Examples

- u($t \pm t_0$), r[n $\pm N_0$], rect($t \pm t_0$), cos[wn $\pm N_0$]



Time Scaling

- Speeding-up/slowing-down videos and audio, rewinding a tape recorder
- Compress or expand the signal in time: *x(at) or x[an]*
  - $|a| < 1$: expand or upsampling
  - $|a| > 1$: compress or downsampling
- Anchored at $t = 0$
- Time reversal: $a < 0$

Examples

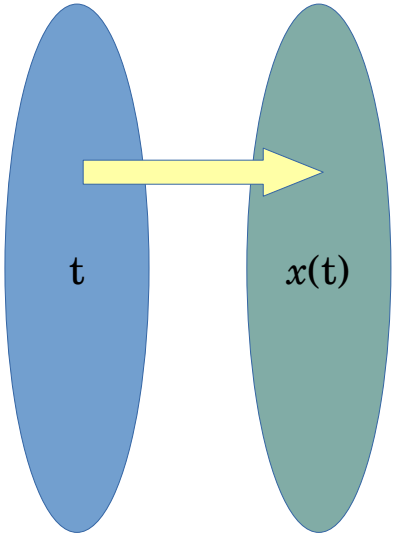
- $\mathrm{u}(at)$, $\mathrm{r}[at]$, $\cos(awt)$

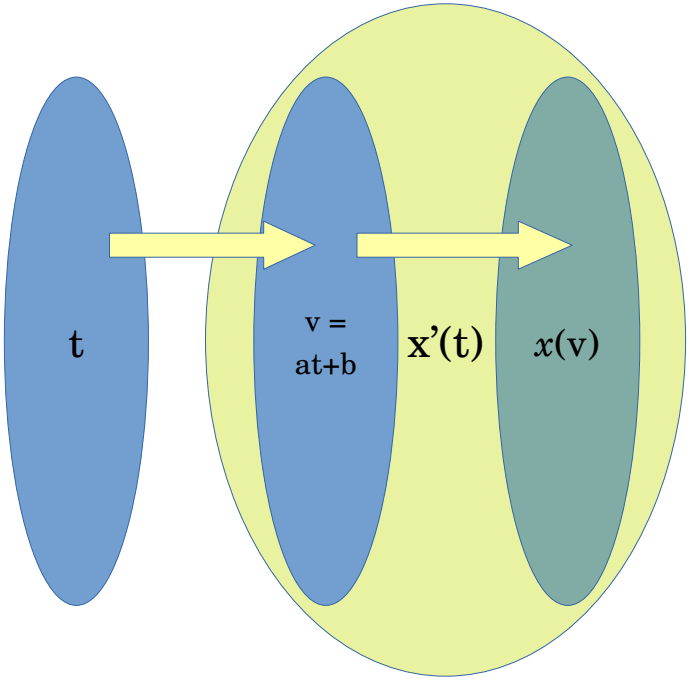
- $\mathrm{rect}(t/T)$

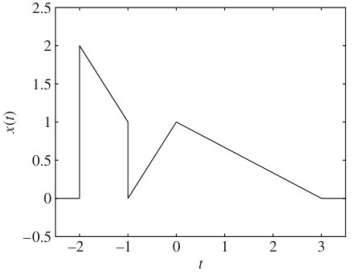
Time Shifting and Scaling

- All together: $y(t) = x(at + b)$ or $y[n] = x[an + b]$
  - Shift by b and then scale by a : $x(t + b)$ and then $x(at + b)$
  - Scale by a and then shift by b/a : $x(at)$ and then $x(a(t + b/a))$
- Order of operations: $x(f_1(f_2(t)))$
  - Reverse operations: $x(f_2^{-1}(f_1^{-1}(f_1(f_2(t)))))$
    - Shift by $-b/a$ and then scale by $1/a$: $y(t - b/a)$ and then $y(t/a - b/a)$
    - Scale by $1/a$ and then shift by $-b$: $y(t/a)$ and then $y((t - b)/a)$

Manipulation of Mapping





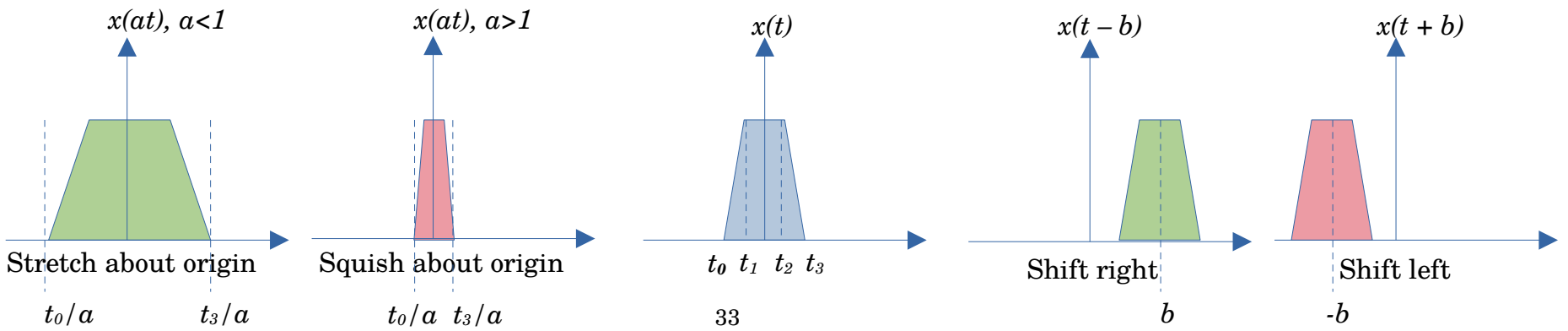


Examples

- $\mathrm{u}(at + b)$, $\mathrm{r}(at + b)$, $\mathrm{rect}[a\mathrm{n} + b]$, $\cos(awt + b)$

Summary of Time Shift and Scale

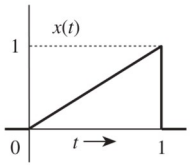
- To compute $x(at + b)$
  - Shift by b (move origin to b) and then scale by a (squish/stretch about the y-axis)
  - Scale by a (squish/stretch about the y-axis) and shift by b/a (move origin to b/a)
  - Note critical points t_0, t_1, …, t_N, compute $(t_0 - b)/a$, $(t_1 - b)/a$, …, $(t_N - b)/a$ and plot

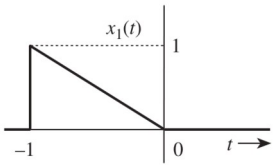


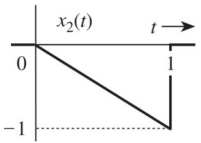
Summary of Time Shift and Scale

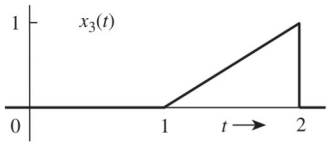
- To get $x(t)$ from $x(at + b)$
  - Shift by $-b/a$ and then scale by $1/a$
  - Scale by $1/a$ and then shift by $-b$
- Given $x(t)$ and $x(at + b)$ with unknown a and b; to find a and b
  - Pick critical points such that $x(av_1 + b) = x(t_1)$ and $x(av_2 + b) = x(t_2)$
  - Now, $a = (t_2 - t_1)/(v_2 - v_1)$ and $b = -av_0$
    - v_0 is the shifted origin
- Similarly for discrete-time signals as well

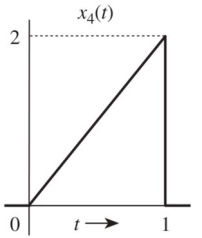
Examples



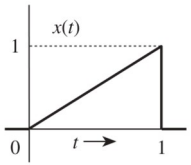




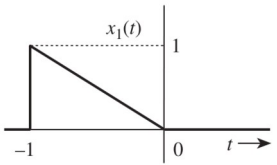




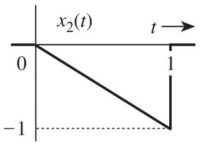
Examples



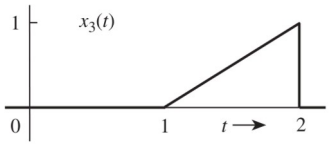
x(t)



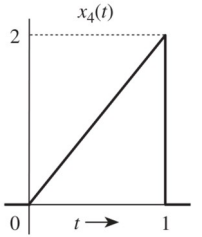
x(-t)



-x(t)

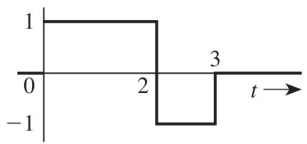


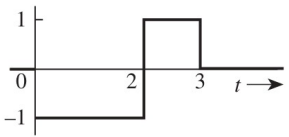
x(t – 1)

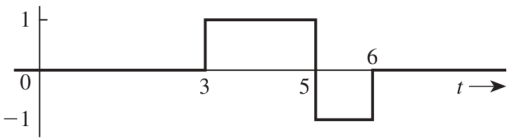


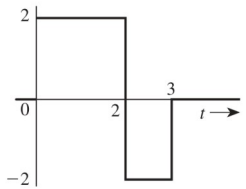
2x(t)

Examples

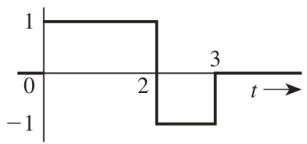




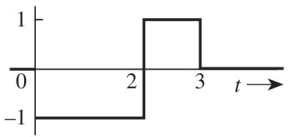




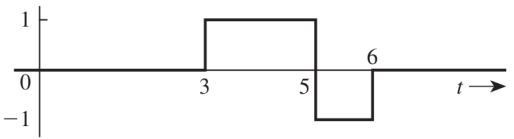
Examples



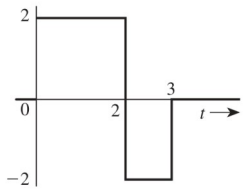
x(t)



-x(t)



x(t – 3)



2x(t)

Examples

- Sketch $x(t)$, $y(t) = x(t/2 - 1)$, retrieve $x(t)$ from $y(t)$

$$x(t) = \begin{cases} 1+t & \text{if } -1 \le t < 0 \\ 1 & \text{if } 0 \le t \le 1 \\ 0 & \text{else} \end{cases}$$

$t \to 2t+2$

$y(2t+2) = x(t)$

- Solution: $x(t) = y(2t + 2)$

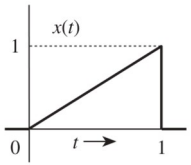
Examples

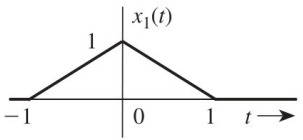
- Sketch u[–3n + 1] and rect[2n - 3]

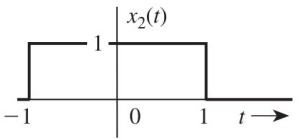
Adding/Subtracting Signals

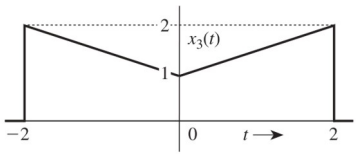
- Linear combination: $z(t) = A\,x(t) + B\,y(t), \quad z[n] = A\,x[n] + B\,y[n]$
  - Add/subtract pointwise (for each value of time)
- Example
  - $\text{Signum}(t) = 2\,u(t) - 1 = u(t) - u(-t)$
  - $\text{rect}(t) = u(t + \tfrac{1}{2}) - u(t - \tfrac{1}{2})$
  - $u[n + 5] - u[n - 5]$

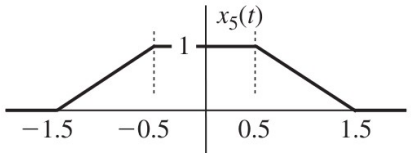
Examples



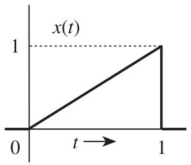




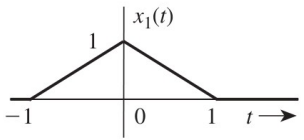




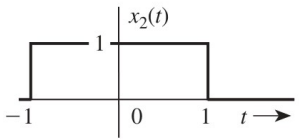
Examples



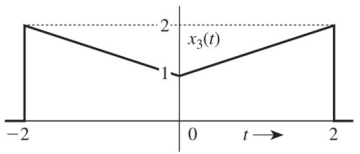
$x(t)$



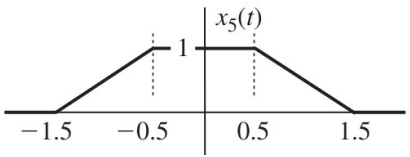
$x(t + 1) + x(-t+1)$



$x(t/2 + ½) + x(-t/2 + ½)$



$x_2(t/2) + x(t/2) + x(-t/2)$



$x_2(2t) + x(t + 1.5) + x(-t + 1.5)$

Examples

- Express tri(t) as integral of some combination of rect(t)

$$1 - \left| \int \text{rect}(t/2)\, dt \right|$$

- $\cos(wt) = 0.5(e^{jwt} + e^{-jwt})$

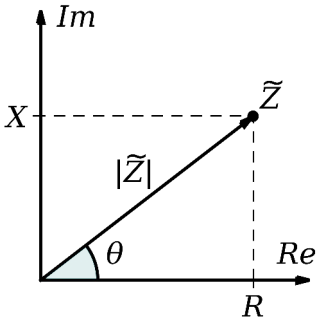
Complex Signals

Signals with Magnitude and Phase

- Audio equalizer: Do all the filters give in-phase output? How to model and analyze this?
- Impedance of a component (Z) = resistance + reactance, $V = IZ$
  - Resistance: opposes flow of current along voltage (always in phase)
  - Reactance: opposes the rate of change of current or voltage (always out of phase)
    - Capacitance: $I = C\,\mathrm{d}V/\mathrm{d}t$ opposes change of voltage (for fixed I) => voltage lags
    - Inductance: $V = \mathrm{d}I/\mathrm{d}t\,L$ opposes change of current (for fixed V) => current lags
- Represent magnitude and phase with complex-valued signals

Phasors

- Complex exponential to represent phase and magnitude
- $Z = Re^{jwt} = R(\cos(wt) + j\sin(wt)) = e^{\sigma + jwt}$ $R = e^{\sigma}$
  - Real part (horizontal component) is cosine
  - Imaginary part (vertical-component) is sine
- Complex sinusoid: Unit magnitude phasor
  - Angular frequency: $w = 2\pi/T$, linear frequency: $f = w/2\pi = 1/T$, periodic with period: T



AC-Analysis of RC-Circuit

$$V_R(t) + V_C(t) = V_0 \cos \omega t$$

$$I(t)R + \frac{1}{C}\int I(t)dt = V_0 \cos \omega t$$

Solving this, we get, $\tan \phi = 1/RC\omega$,

$$V_R(t) = V_0 \frac{RC\omega}{\sqrt{1 + (RC\omega)^2}} \cos(\omega t + \phi)$$

$$V_C(t) = V_0 \frac{1}{\sqrt{1 + (RC\omega)^2}} \sin(\omega t + \phi)$$

AC-Analysis of RC-Circuit – with Phasors

$$V_R(t) + V_C(t) = I(t)R + \frac{1}{C}\int I(t).dt = V(t)$$

$$\boxed{V(t) = V_0 e^{j\omega t};} \quad I(t) = ae^{j\omega t} \Rightarrow a = V_0 \frac{jC\omega}{1 + jRC\omega}$$

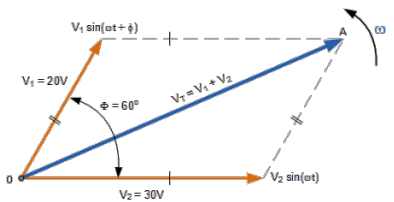
$$V_R(t) = V_0 \frac{jRC\omega}{1 + jRC\omega} e^{j\omega t} = V_0 \frac{RC\omega}{\sqrt{1 + (RC\omega)^2}} e^{j\omega t + j\phi}$$

$$V_C(t) = V_0 \frac{1}{1 + jRC\omega} e^{j\omega t} = V_0 \frac{1}{\sqrt{1 + (RC\omega)^2}} e^{j\omega t + j\phi - j\frac{\pi}{2}} \qquad (\tan^{-1}\theta + \tan^{-1}\frac{1}{\theta} = \frac{\pi}{2})$$

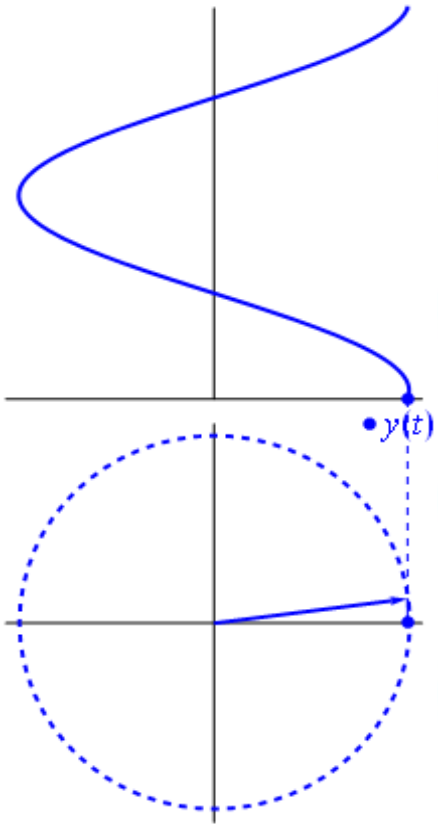
Phasors are eigenfunctions of integration and differentiation

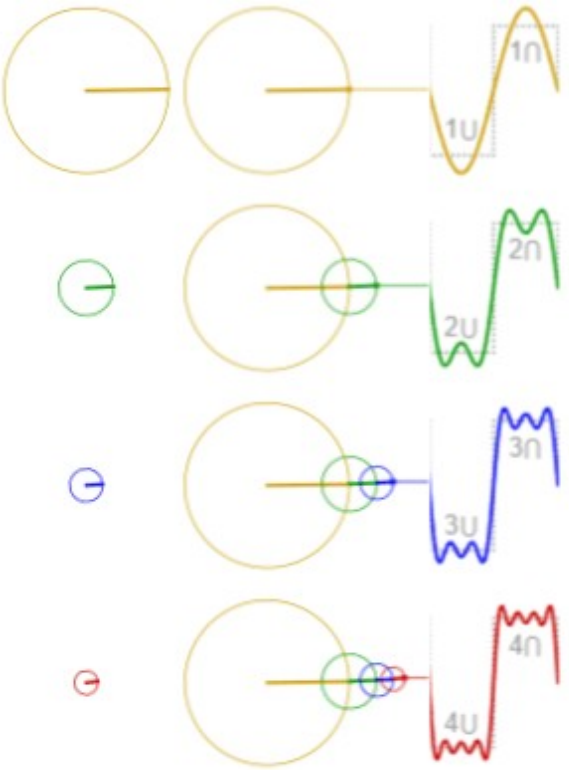
Multiplying/Adding Phasors

- Product of phasors: $re^{jwt}\ qe^{jvt} = (e^{a+jwt})(e^{b+jvt}) = e^{(a+b)}\,e^{j(w+v)t}$
  - Resultant is also a phasor (sinusoid)
  - Frequencies/phases add-up and magnitudes get multiplied
- Sum of phasors: $(e^{a+jwt}) + (e^{b+jvt})$
  - Resultant is not a phasor (sinusoid) if $w \neq v$
  - Magnitude = $\sqrt{(\ r^2 + q^2 + 2rq\cos(wt - vt)\ )}$
  - Phase = $\tan^{-1}(\ (r\sin(wt) + q\sin(vt)) / (r\cos(wt) + q\cos(vt))\ )$

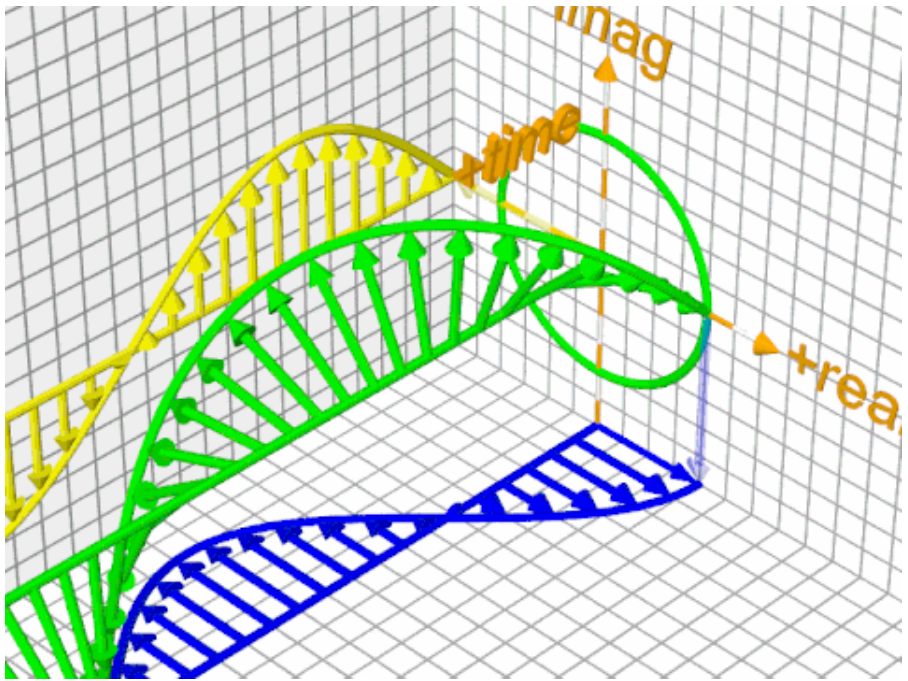


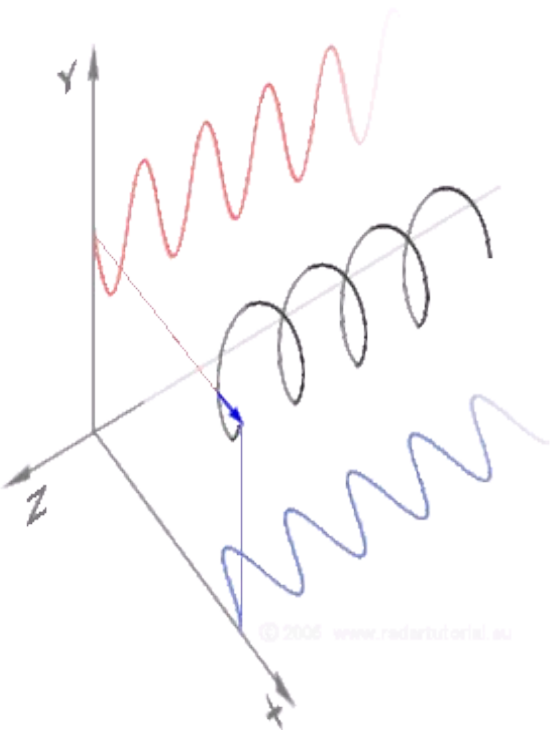
Sum of Phasors





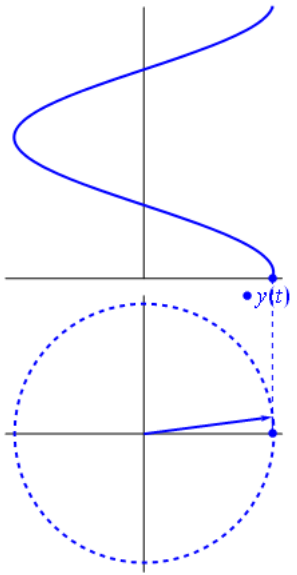
Phasors





Negative Frequency

- Sign of frequency: direction of rotation of the phasor
  - Positive is anticlockwise
  - Negative is clockwise
- Frequency: rate of change of phase
  - Positive is increasing phase
  - Negative is decreasing phase
- Negative frequency (mathematically) quantifies the above for complex sinusoid



Visualizing Complex-valued Signals

- $x(t)$ – at each time instant, can be complex valued
  - $x(t) = x_r(t) + j\, x_i(t)$
    - Sum of two real-valued signals multiplied by appropriate complex constants
- Graphical representation of complex signals
  - Real and imaginary parts as functions of time: $x(t) = x_r(t) + j\, x_i(t)$
  - Magnitude and phase as functions of time: $x(t) = |x(t)| \exp(j \angle x(t))$
    - Phase wrap: between $-\pi$ to $+\pi$

Complex-valued Signals – Examples

- Example: sketch the real, imaginary, magnitude and phase plots of $x(t) = e^{\sigma t + jwt}$
  - $x_r(t) = e^{\sigma t}\cos(wt)$
  - $x_i(t) = e^{\sigma t}\sin(wt)$
  - $\sigma > 0$: Growing sinusoid ; $\sigma < 0$: Decaying sinusoid
- Examples: $x(t) = \cos(wt)$
  - Phase: 0 for positive values of $x(t)$ and π for negative values of $x(t)$

Summary of Phasors

- Phasors (complex exponentials) are eigenfunctions of integration and differentiation
  - This simplifies analysis of phases (phase lag/lead) in circuits
  - A unit magnitude phasor is a complex sinusoid
  - The direction of rotation determines the sign of frequency
    - Anticlockwise is positive; clockwise is negative
  - Requires two plots for visualization (magnitude and phase)
    - Phase wrapping in phase plot so that phases are between $-\pi$ and $+\pi$

Phasors - Example

- If the input of a series RC circuit is an AC voltage given by $x(t) = 5\cos(2\pi \times 50t)$, sketch the voltage across the resistor (2 kΩ) and capacitor (16 uF).

$$V_R(t) + V_C(t) = I(t)R + \frac{1}{C}\int I(t).dt = V_0 e^{j\omega t}$$

$$V_R(t) = V_0 \frac{|RC\omega|}{\sqrt{1+(RC\omega)^2}} e^{j\omega t + j\phi}, \quad \tan\phi = 1/RC\omega$$

$$V_C(t) = V_0 \frac{1}{\sqrt{1+(RC\omega)^2}} e^{j\omega t + j(\phi - \frac{\pi}{2})}$$

Phasors - Example

Properties of Signals

Even/Odd

- Even signal: $x(t) = x(-t)$ or $x[n] = x[-n]$
  - Symmetric about t = 0
  - Area under even signal is $\int_{-\infty}^{\infty} x(t).dt = 2\int_{0}^{\infty} x(t).dt$
- Odd signal: $x(t) = -x(-t)$ or $x[n] = -x[-n]$
  - Anti-symmetric about t = 0
  - Area under odd signal is zero

Even/Odd - Examples

- $\cos(wt)$, $\sin[wn]$, and $\text{rect}(t - ½)$

Even/Odd - Examples

- e^{-t} for t > 0, and $x(t) = \begin{cases} 1+t & \text{if } -1 \le t < 0 \\ 1 & \text{if } 0 \le t \le 1 \\ 0 & \text{else} \end{cases}$

Even/Odd

- Signal decomposition: $x(t) = x_{even}(t) + x_{odd}(t)$

- $x_{even}(t) = \frac{1}{2}\left(x(t) + x(-t) \right)$ or $x_{even}[n] = \frac{1}{2}\left(x[n] + x[-n] \right)$
- $x_{odd}(t) = \frac{1}{2}\left(x(t) - x(-t) \right)$ or $x_{even}[n] = \frac{1}{2}\left(x[n] - x[-n] \right)$

- The even and odd components are orthogonal to one another
  - Orthogonal signals: $\int_{-\infty}^{\infty} x^*(t)y(t)dt = \int_{-\infty}^{\infty} x(t)y^*(t)dt = 0$
    - Equivalent to dot-product for continuous-time signals (summation for DT)
    - For real-valued signals: $\int_{-\infty}^{\infty} x(t)y(t)dt = 0$

Orthogonal Signals - Examples

- Check if the following are orthogonal
  - rect(t - 1) and rect(t + 1)
    - Signals whose non-zero portions do not overlap are orthogonal
  - rect[t] and tri[t]

Orthogonal Signals - Examples

- Check if the following are orthogonal
  - $\cos(wt)$ and $\cos(vt)$
  - $\sin(wt)$ and $\cos(vt)$
  - e^{jwt}, e^{-jwt}

Orthogonal Signals - Examples

- Check if the following are orthogonal
  - cos(wt) and cos(vt)
    - Yes $w \neq v$
  - sin(wt) and cos(vt)
    - Yes
  - e^{jwt}, e^{-jwt}
    - Yes

Even/Odd - Examples

- Decompose to even/odd parts: rect(t – ½), e^{-t} for t > 0, and $x(t) = \begin{cases} 1+t & \text{if } -1 \le t < 0 \\ 1 & \text{if } 0 \le t \le 1 \\ 0 & \text{else} \end{cases}$

Combination of Even/Odd Signals

$x(t)$	$y(t)$	$x(t).y(t)$	$x(t) + y(t)$
Odd	Even	Odd	Neither
Even	Even	Even	Even
Odd	Odd	Even	Odd

- Examples: $\tan(wt)$, e^{jwt}, e^{-jwt}

Conjugate Symmetry

- Symmetry for complex signals
  - Conjugate symmetric signal: $x(t) = x^*(-t)$ or $x[n] = x^*[-n]$
  - Conjugate asymmetric (skew-symmetric) signal: $x(t) = -x^*(-t)$ or $x[n] = -x^*[-n]$
- Signal decomposition: $x(t) = x_{cs}(t) + x_{ca}(t)$
  - $x_{cs}(t) = \frac{1}{2}\left(x(t) + x^*(-t) \right)$ or $x_{cs}[n] = \frac{1}{2}\left(x[n] + x^*[-n] \right)$
  - $x_{ca}(t) = \frac{1}{2}\left(x(t) - x^*(-t) \right)$ or $x_{ca}[n] = \frac{1}{2}\left(x[n] - x^*[-n] \right)$
- Unlike even/odd, CS/CA does not affect area (and so, orthogonality)

Conjugate Symmetry and Even/Odd

- For real signals, CS = Even and CA = Odd
- For a complex signal $x(t) = x_r(t) + j\, x_i(t)$
  - Conjugate symmetry implies
    - $x_r(t)$ is even and $x_i(t)$ is odd
  - Conjugate asymmetry implies
    - $x_r(t)$ is odd and $x_i(t)$ is even
  - $|x(t)|$ is even and $\sphericalangle x(t)$ is odd (but phase can be neither due to wrapping and $\tan^{-1}$ computation based on quadrant)

Conjugate Symmetry - Examples

- e^{jwt}, e^{-jwt}
  - For a phasor, change of sign of t and conjugation are equivalent
    - Reflection about the real axis
  - CS but neither even nor odd
- $\exp(jwt^2)$: Even but neither CS nor CA
- $\cos(wt)$

Combination of CS/CA Signals

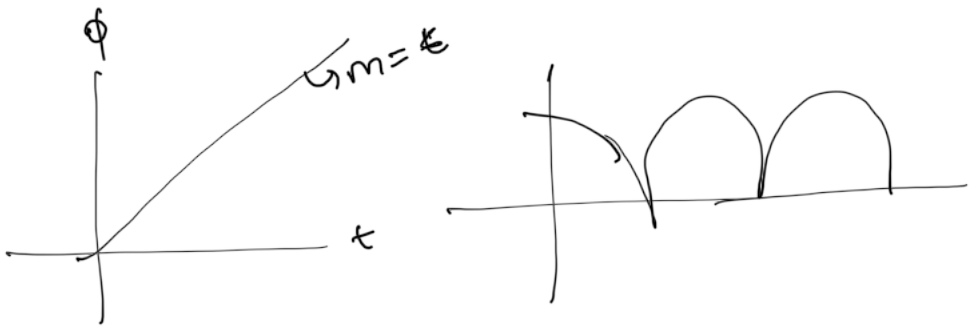
$x(t)$	$y(t)$	$x(t).y(t)$	$x(t) + y(t)$
CA	CS	CA	Neither
CS	CS	CS	CS
CA	CA	CS	CA

- Examples: $(e^{jwt} + e^{-jwt})/2j$

Conjugate Symmetry - Example

- Sketch real/imaginary and magnitude/phase plots of $\cos(wt)\, e^{jwt}$

$\cos wt + j \sin wt$

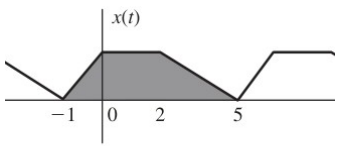


Conjugate Symmetry - Example

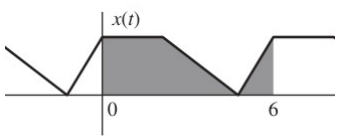
- Sketch real/imaginary and magnitude/phase plots of $e^{\sigma t + jwt}$

Periodicity

- Periodic signal: A signal whose values repeat after a period T
  - Period: The smallest value of T (or N) > 0 s.t. $x(t + T) = x(t)$ or $x[n + N] = x[n]$
    - T (or N) – fundamental period; nT (or mN) – also periods of repetition
  - Area in any interval of width nT is the same



(a)



$$\int_a^{a+nT} x(t)dt = \int_b^{b+nT} x(t)dt$$

Periodicity

- Aperiodic signal: A signal whose values do not repeat at any interval
  - All finite length signals are aperiodic
  - Practically, all signals are finite length
- Analytically, periodic signals necessarily extend from -∞ to +∞
- In real-life, we measure periodicity within our time window of interest

Combination of Periodic Signals

- $x(t)$ has period T_1, $y(t)$ has period T_2
  - $T = nT_1 = mT_2$
    - ~~T_1/T_2 has to be rational~~
  - $x(t)$. $y(t)$ is periodic with the least common integer multiple of T_1 and T_2
  - $x(t) + y(t)$ is periodic with the least common integer multiple of T_1 and T_2
- Examples: $\sin(t) + \cos(\pi t)$, $e^{-j3t} + \tan(2t)$
- For $x[n]$ with period N_1, $y[n]$ with period N_2, $x[n] + y[n]$ has a period = LCM(N_1, N_2)

Time Scaling/Shifting of Periodic Signals

- Time shifting does not affect periodicity
- Time scaling changes the period
  - If $x(t)$ is periodic with period T, then the $x(at)$ is periodic with period $T/|a|$
    - For $|a| < 1$, period increases
    - For $|a| > 1$, period decreases
- Similarly for DT signals

Periodic Signals - Examples

- Find periodicity of sin(t), sin[n], cos(πt), cos[πn], rect(t), e^{wt}, e^{-jwt}

- $\sum_{n=-\infty}^{\infty} \mathrm{rect}(t - nT)$

Periodic Signals - Examples

- Find periodicity of sin(t), cos(πt), rect(t), e^{wt}, e^{-jwt}
  - 2π, 2, aperiodic, aperiodic, $2\pi/w$

- $\sum_{n=-\infty}^{\infty} \text{rect}(t - nT)$
  - For T > 1, period = T

Energy of Signals

- Energy is area under the signal magnitude squared
  - $E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$ or $\sum_{n=-\infty}^{\infty} |x[n]|^2$
- If $x(t)$ is the current in an 1 Ohm resistor, then E_x is the heat dissipated in it
- Represents squared-norm (like length) of the signal than actual physical energy
  - Conservation of energy is not true: $E_x + E_{-x} \neq 0 = E_{x-x}$ (triangle inequality)
  - However Pythagorean relation is true (as E_x is like squared-length)
- Signals with finite energy ($E_x < \infty$) are called energy signals

Energy of Signals - Examples

- Find energy of rect(t), tri[n], and $x(t) = \begin{cases} 1+t & \text{if } -1 \leq t < 0 \\ 1 & \text{if } 0 \leq t \leq 1 \\ 0 & \text{else} \end{cases}$

Energy of Signals - Examples

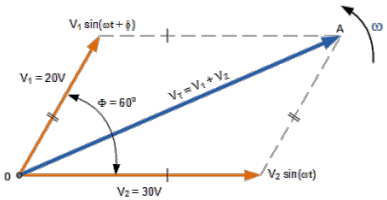
- Find energy of e^{-t} for t > 0

Energy of Orthogonal Signals

- Energy of sum of signals (for both CT and DT)

$$\int_{-\infty}^{\infty} |x(t)+y(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2.dt + \int_{-\infty}^{\infty} |y(t)|^2 dt + \int_{-\infty}^{\infty} x^*(t)y(t)dt + \int_{-\infty}^{\infty} x(t)y^*(t)dt$$

  - $E_x + E_y = E_{x+y}$ only when $x(t)$ and $y(t)$ are orthogonal
    - $\int_{-\infty}^{\infty} x^*(t)y(t)dt = \int_{-\infty}^{\infty} x(t)y^*(t)dt = 0$
    - Follows from length analogy



- Also, $E_{xy} \neq E_x E_y$ (true only if $x(t) = cy(t)$)

Energy of Orthogonal Signals - Examples

- Find the energy of $\sum_{n=-2}^{1} \text{rect}(t - nT)$

Power of Signals

- Some signals may have infinite energy
  - Examples: u(t), all periodic signals: cos(wt), e^{wt} (non-energy signals)
- Power of a signal is defined as
  - $P_x = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$
  - Average energy of the signal
  - If $E_x < \infty$ then $P_x = 0$, and $P_x > 0$ then $E_x = \infty$
  - Signals with finite power ($P_x < \infty$) are called power signals

Power of Signals - Examples

- Find power of rect(t), tri(t), e^{-t} for t > 0 and $x(t) = \begin{cases} 1+t & \text{if } -1 \le t < 0 \\ 1 & \text{if } 0 \le t \le 1 \\ 0 & \text{else} \end{cases}$

Power of Signals - Examples

- Find power of $A\cos(wt)$, $A\sin(wt) + B\sin(vt)$, u(t)
- Find power of $A\cos[wn]$, u[n]

Power of Signals - Examples

- Find power of e^{at} for a positive, negative and imaginary

Power of Periodic Signals

- Average energy in one period is $E_{x,T} / T = \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 . dt$
  - For periodic signals, energy in every period is the same
  - Power is average energy over time
    - Power of periodic signals = average energy in one period

Effect of Time Scale/Shift on Energy/Power

- Time shifting does not affect energy/power
- Time scaling
  - If energy of $x(t)$ is E_x then the energy of $x(\alpha t)$ is $E_x / |\alpha|$
    - As signal stretches or compresses, the area increases or decreases, respectively
    - Example: rect(αt), rect[αn]
  - Power remains same
    - As the time-average (over $-\infty$ to $+\infty$) does not change

Delta Function

Dirac Delta Signal

- How to model the instantaneous jumps in current/voltages?
  - What is the current across a capacitor when a step voltage is applied?
  - What is the derivative of step function?
  - Dirac delta signal: $\delta(t) = du(t)/dt$
    - Zero for all values of time except $t = 0$
    - Area is 1, $\int \delta(t)\, dt = 1$ (infinitesimally small in time, infinitely large in amplitude)
    - Not a true function

Dirac Delta Signal - Properties

- Even signal
- Time scaling is scaling of area: $\delta(at) = \delta(t)/|a|$
- $du(-t)/dt = -\delta(t)$
- Sifting property
  - $\delta(t)x(t) = x(0)\delta(t)$
  - $\delta(t-a)x(t) = x(a)\delta(t-a)$ $\quad \delta(t-a)\,x(t-b) = x(a-b)\,\delta(t-a)$
  - $\int \delta(t)x(t)\,dt = x(0)$

Dirac Delta Signal - Examples

- Find the derivative of *A*rect(t) and $x(t) = \begin{cases} 1+t & \text{if } -1 \le t < 0 \\ 1 & \text{if } 0 \le t \le 1 \\ 0 & \text{else} \end{cases}$

- Find the integral of δ(t) in (-∞, t) $= u(t)$

Dirac Delta Signal - Examples

- Find the derivative of $\sum_{n=-\infty}^{\infty} \text{rect}(t - nT)$

- Plot $\sum_{n=-\infty}^{\infty} x(t)\delta(t - nT)$
  - A CT representation of a DT signal

Kronecker Delta Signal

- The discrete time delta function: Kronecker delta
  - $\delta[n] = 1$ if $n = 0$, 0 elsewhere
  - $\delta[n] = u[n] - u[n-1]$
  - $\delta[n] = 1$
  - $\delta[n-a]x[n-b] = x[a-b]\delta[n-a]$
  - $\Sigma_n \delta[n]x[n] = x[0]$

Grand Summary of Signals

Summary

- Continuous-time signals (e.g., u(t), δ(t), rect(t), cos(wt), e^{jwt}, …)
  - Time and amplitude are from a continuum, can model voltage/current, etc.
  - Negative time represents signal value before initialization
  - Signal from –∞ to +∞ in time is for analytical convenience
    - Real-world signals can be considered to be $x(t)\ \mathrm{rect}(t/T_1 - ½ - T_0/T_1)$
- Time shift: Move the signal to left $x(t + a)$ or right $x(t - a)$
- Time scale: Stretches ($a < 1$) or compresses ($a > 1$) the signal $x(t)$

Summary

- Discrete-time signals (e.g., u[n], δ[n], rect[n], cos[wn], e^{jwn}, …)
  - Time is discrete and amplitude is from a continuum, forms a sequence
  - Sinusoids with irrational frequency coefficients are aperiodic
  - Time variable denotes just the index of a sample
    - Sampling time: time difference between any two consecutive samples is always the same
- Time shift: Move the signal to left x[n + N] or right x[n – N]
- Time scale: Stretches (a < 1) or compresses (a > 1); x[an]

Summary

- Time scaling: to compute $x(at + b)$ or $x[an + b]$
  - Shift by b (move origin to b) and then scale by a (squish/stretch about the y-axis)
  - Scale by a (squish/stretch about the y-axis) and shift by b/a (move origin to b/a)
- Phasors are eigenfunctions of integration/differentiation – they are complex sinusoids
  - Negative frequency is clockwise rotation of phasor
- Complex-valued signals need two plots: magnitude/phase or real/imaginary
- Symmetries: even, odd, conjugate symmetry and conjugate asymmetry

Summary

- Periodic/aperiodic signals; angular frequency = 2π/period rad/s; discrete sinusoids
- Energy of a signal is the area under the signal-squared (squared-length of vector)
- Power of a signal is the asymptotic time average of its energy
  - All practical signals are aperiodic and energy signals
    - But theoretical approximations help in analysis
- Orthogonal signals: component of one signal in another is zero
  - Real (or complex)-valued sinusoids of different frequencies are orthogonal

Summary

	Sum	Product	Time Shift	Time Scale
Even/Odd	Depends	Depends	Changes	Remains same
Conjugate Sym/Asym	Depends	Depends	Changes	Remains same
Periodicity	Periodic only if T_1/T_2 is rational	Periodic only if T_1/T_2 is rational	Remains same	$T/\lvert a \rvert$
Energy	Adds up only for $x(t) \perp y(t)$	Multiplies only for $x(t) = c\, y(t)$	Remains same	$E_x/\lvert a \rvert$
Power	Adds up only for $x(t) \perp y(t)$	Multiplies only for $x(t) = c\, y(t)$	Remains same	Remains same